Namit Anand

Curriculum Vitæ

University of Southern California
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Research interests

Quantum resource theories (specifically, quantum coherence), open quantum systems and non-Markovian processes, application of quantum information-theoretic tools to many-body theory (specifically, many-body localization and quantum chaos), and understanding the role of coherence and entanglement in quantum algorithms.

Education

2017 – present **Doctor of Philosophy, Physics**, *University of Southern California*, Los Angeles, CA, USA. *Advisor*: Paolo Zanardi. *Co-advisors*: Aaron Lauda and Todd A. Brun.

2016 – 2017 **Visiting Student Researcher, Physics**, *Harish-Chandra Research Institute*, Prayagraj, India.

2011 – 2016 **Integrated Master of Science (BS+MS), Physics**, *National Institute of Science Education and Research*, Bhubaneswar, India.

Publications & Preprints

1. **Information Scrambling and Chaos in Open Quantum Systems**, *Paolo Zanardi and Namit Anand*, arXiv:2012.13172 [quant-ph] (2020).
2. **Quantum coherence as a signature of chaos**, *Namit Anand, Georgios Styliaris, Meenu Kumari, and Paolo Zanardi*, arXiv:2009.02760 [quant-ph] (2020).
3. **Information Scrambling over Bipartitions: Equilibration, Entropy Production, and Typicality**, *Georgios Styliaris, Namit Anand, and Paolo Zanardi*, arXiv:2007.08570 [quant-ph] (2020).
4. **Quantum coherence and the localization transition**, *Georgios Styliaris, Namit Anand, Lorenzo Campos Venuti, and Paolo Zanardi*, Phys. Rev. B **100**, 224204 (2019), [arXiv:1906.09242].
5. **Quantifying non-Markovianity: a quantum resource-theoretic approach**, *Namit Anand and Todd A. Brun*, arXiv:1903.03880 [quant-ph] (2019).
6. **Demonstration of fidelity improvement using dynamical decoupling with superconducting qubits**, *Bibek Pokharel, Namit Anand, Benjamin Fortman, and Daniel A. Lidar*, Phys. Rev. Lett. **121**, 220502 (2018).
   – Popular summary of our work in Phys.org, ScienceDaily, and the Scientific American.
   – Featured on the Rigetti website.
7. **Asymmetry and coherence weight of quantum states**, *Kaifeng Bu, Namit Anand, and Uttam Singh*, Phys. Rev. A **97**, 032342 (2018).

8. **Coherence and Entanglement Monogamy in the Discrete Analogue of Analog Grover Search**, *Namit Anand and Arun K. Pati*, arXiv:1611.04542 [quant-ph] (2016).
9. **Comment on "Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics"**, *Namit Anand*, Eur. J. Phys. **37** 048003 (2016).
10. **Do quantum strategies always win?**, *Namit Anand and Colin Benjamin*, Quantum Information Processing **14** (11), 4027-4038 (2015).

⋆ [Google Scholar] and [arXiv].

Relevant Graduate Coursework

- Introduction to Quantum Information Processing (*Course Instructor*: Prof. Todd A. Brun)
- Quantum Error Correction (*Course Instructor*: Prof. Daniel A. Lidar)
- Theory of Open Quantum Systems (*Course Instructor*: Prof. Daniel A. Lidar)
- Quantum Information Theory (*Course Instructor*: Prof. Todd A. Brun)
- Quantum Information Science and Many-body Physics (*Course Instructor*: Prof. Paolo Zanardi)
- Quantum Spin Chains (*Course Instructor*: Prof. Hubert Saleur)

Teaching Experience

(Fall '17 & Spring '18) Graduate Teaching Assistant for Physics 135aL: Physics for the Life Sciences. Student Evaluation Rating: 5.0/5.0

Awards and Fellowships

Summer 2021 Recipient of the USC Gold Family Fellowship

Award: $5,000 (USD)

Summer 2018 Recipient of the Quantum computing fellowship at the Los Alamos National Laboratory.

Award: $12,200 (USD)

2017 – 2022 University of Southern California, Department of Physics & Astronomy.

Award: $27,500 (USD) annually; full financial support through teaching assistantship and one year of fellowship for completion of the Ph.D. program.

2017 Graduate fellowships declined for USC: University of Maryland; Dartmouth College; University of Waterloo; Stony Brook (MS); Université libre de Bruxelles; Scuola Normale Superiore di Pisa.

Summer 2015 One of the five summer fellows (internationally) for the fully funded I.I.A.S.S. Summer internship program, Italy.

Award: €3,000.

2011 – 2016 Recipient of Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship by the Department of Science and Technology, Government of India from 2011 - 2016.

Award: ₹80,000/year for 5 years.

Undergraduate Research Experience

Aug '16 – May '17	**Identifying Topological Order with Entanglement – the role of quantum information in condensed matter physics**, *Prof. Sumathi Rao*, Harish-Chandra Research Institute.
Sep '16 – May '17	**Concentration of Measure phenomena in quantum information theory**, *Namit Anand and Uttam Singh*, Harish-Chandra Research Institute.
May '16 - '17	**Density Matrix Renormalization Group Algorithms**, *Dr. Anamitra Mukherjee*, NISER.
Aug '15 – '16	**Quantum Games, Walks, and Algorithms**, *Dr. Colin Benjamin*, NISER.
May – July 2015	**Characterizing phases in one-dimensional systems using entanglement**, *Prof. Roberta Citro*, University of Salerno.
Jan – May 2015	**Simulation of two-level quantum system using classical coupled oscillators**, *Dr. Ashok Mohapatra*, NISER.

Computational Skills

Programming Languages	C, C++, Python
Specialized Software	Mathematica and MATLAB
Markup Languages	LaTeX, HTML, HTML5, PHP
Software adept	Adobe Suite (Photoshop, Audition) and Game Development (UDK)
Numerical packages	*Lapack* for C++ & Python, MATLAB on HPC, *Quantum Mathematica* for quantum computing simulations in Mathematica, and CVX for convex optimization in MATLAB
Quantum simulations	Simulating quantum error-correction on the IBM Quantum Experience

Professional References

Paolo Zanardi

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Todd A. Brun

Professor of Electrical Engineering-Systems, Computer Science, and Physics and Astronomy
University of Southern California
Email: tbrun@usc.edu
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Daniel A. Lidar

(*Viterbi Professor of Engineering*)
Professor of Electrical Engineering-Systems, Chemistry, and Physics and Astronomy
University of Southern California
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Arun K. Pati

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